Data Science 500 Assignment

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QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

Submit the assignment in PDF format on Moodle. Use the specified cover page provided.

Include a signed declaration of originality.

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GIT-HUB LINK

https://github.com/kingmonty202-gif/Data-Science-Programming-Assignments.git

Introduction

This report details a comprehensive end-of-the-end data science study carried out for a publishing analytics team that provides catalog optimization advice to bookstores and online merchants. The objective was to investigate the factors that influence book prices and ratings on Books to Scrape (books.toscrape.com), a sandbox e-commerce website designed for web scraping practice, and to build a predictive model capable of estimating book prices from publicly available catalogue information.

The project employs a typical data science workflow, which includes data collection via online scraping, data preparation and cleaning, exploratory data analysis and visualization, linear regression predictive modeling, and expert reporting of results. A total of 200 novels from various genre categories were gathered and examined. The generated insights are meant to enable retailers and publishers in making data-driven pricing and catalog decisions.

Methodology

2.1 Data Acquisition

Data was acquired by scraping the Books to Scrape website using Python's requests and BeautifulSoup libraries. The catalogue was traversed page by page, with each product card parsed to extract the individual book URL, which was then followed to obtain detailed product information. The following fields were captured for each book:

- Book title
- Price (in GBP)
- Star rating (1–5)
- Stock availability and quantity
- Category
- Product description
- Universal Product Code (UPC)
- Price excluding tax, price including tax, and tax
- Number of customer reviews

A total of 200 books were successfully scraped and stored in a structured CSV file. A short delay was included between requests to avoid overloading the server.

2.2 Data Cleaning and Preparation

The raw scraped data was loaded into a Pandas DataFrame and inspected for quality issues. Duplicate entries were removed based on the UPC field. Missing values were identified (only one book lacked a description) and filled with empty strings. All numeric columns — price, rating, quantity, and tax — were converted to appropriate data types.

Feature engineering was then applied to derive additional variables intended to support modelling, including:

- title_length — number of characters in the title
- description_length — number of characters in the description
- description_word_count — number of words in the description
- has_description — binary flag indicating whether a description exists
- price_band — a categorical banding of prices (Low, Medium, High, Premium)

The cleaned dataset was saved as books_clean.csv and contains 200 rows and 19 columns.

2.3 Exploratory Data Analysis

Exploratory analysis was performed using Matplotlib and Seaborn. Visualisations included:

- Histogram of price distribution
- Countplot of star ratings
- Boxplot of price by star rating
- Bar chart of average price by category
- Correlation heatmap of numeric features
- Boxplot of price by stock availability

These visualisations were used to identify patterns and relationships within the catalogue.

2.4 Predictive Modelling

A Linear Regression model was constructed using Scikit-learn. The target variable was book price, and the features included:

- Numeric: rating, quantity, title length, description length, reviews
- Categorical: category, stock

A preprocessing pipeline was applied using StandardScaler for numeric features and OneHotEncoder for categorical features. The data was split into 80% training and 20% testing sets using a fixed random seed for reproducibility. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

Results

3.1 Catalogue Insights

Price Distribution: Book prices in the catalogue range roughly between £10 and £60, with most prices clustering between £20 and £40. The distribution is right-skewed, indicating a small number of higher-priced titles.

Ratings Distribution: Star ratings are relatively balanced across the 1–5 scale, with no single rating dominating the catalogue.

Price by Rating: The boxplot shows that star rating has only a weak effect on price. Books of all rating levels span a similar price range, suggesting that customer perception (as measured by stars) is not strongly linked to pricing.

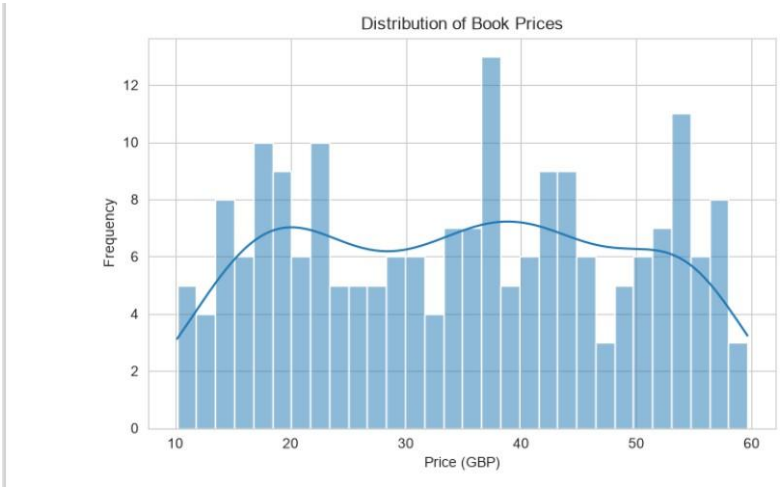
Average Price by Category: Significant variation exists across genres. Categories such as Autobiography and Philosophy have lower average prices, while categories such as Christian, Science, and Self Help show higher averages.

Correlation Matrix: Most numeric features show weak correlations with price. This is an early indication that the linear model may have limited predictive power.

Price by Stock: Books that are in stock versus out-of-stock show similar price distributions, indicating stock status is not a strong price driver.

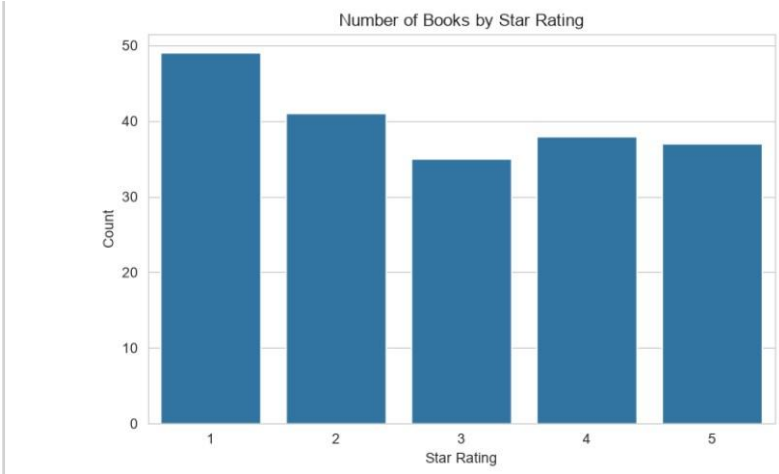
3.1.1 Figure 1

Distribution of Book Prices in the Catalogue. This histogram shows the frequency of book prices (in GBP) across the 200 scraped titles, illustrating a right-skewed distribution.



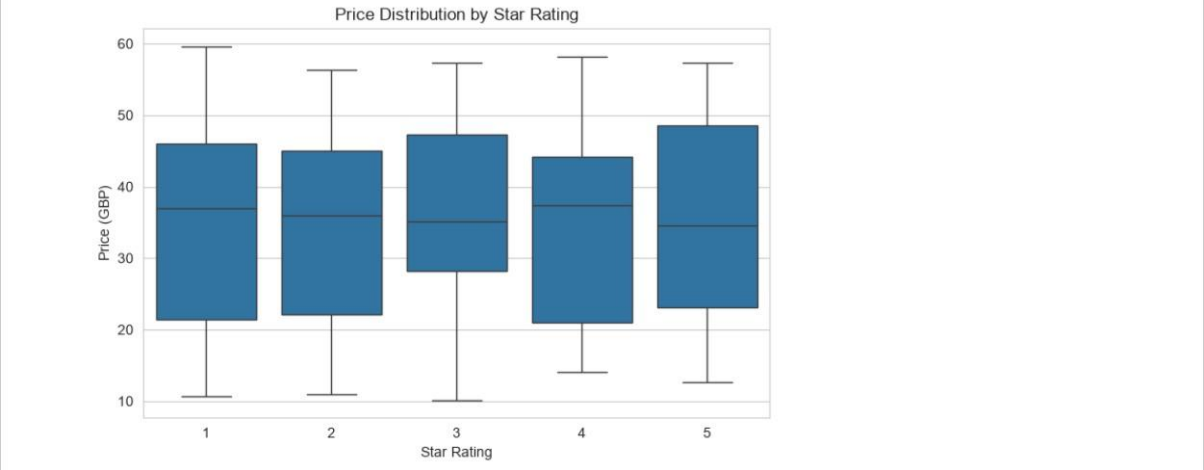
3.1.2 Figure 2

Frequency of Books by Star Rating. This bar chart displays the number of books falling into each star rating category (1 to 5), showing a relatively balanced distribution across ratings.



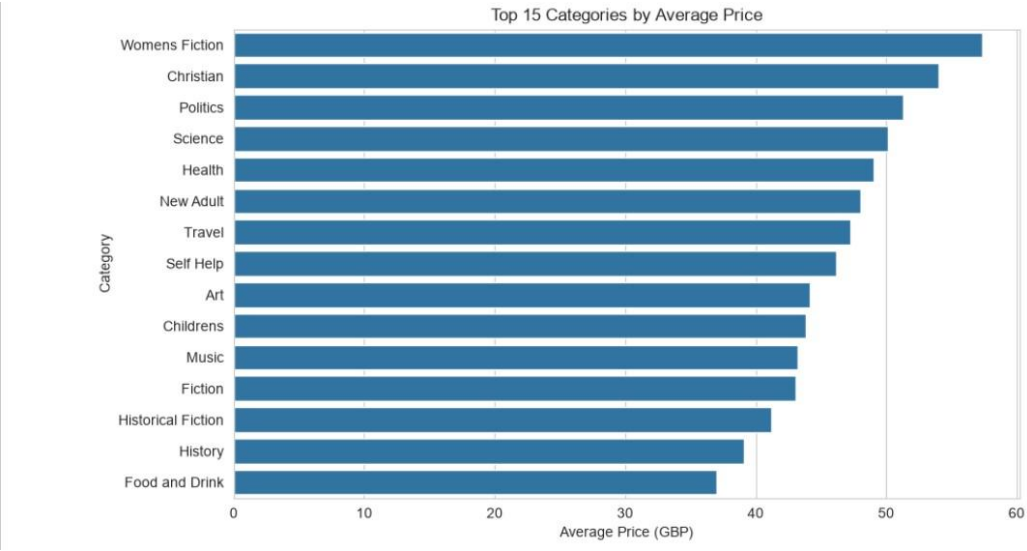
3.1.3 Figure 3

Relationship Between Book Price and Star Rating. This boxplot illustrates the spread and median price of books across the five-star ratings, revealing a weak correlation between rating and price.



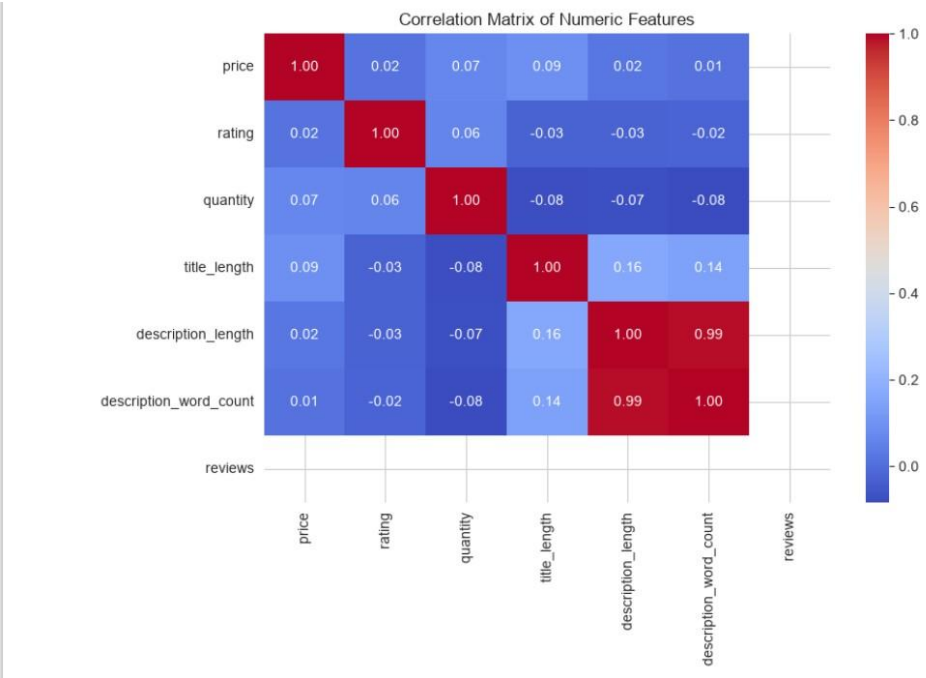
3.1.4 Figure 4

Top 15 Categories by Average Book Price. This horizontal bar chart highlights the genre categories with the highest average book prices, demonstrating significant variation across genres.



3.1.5 Figure 5

Correlation Matrix of Numeric Features. This heatmap shows Pearson correlation coefficients between key numeric variables, indicating weak linear relationships between price and the extracted features.



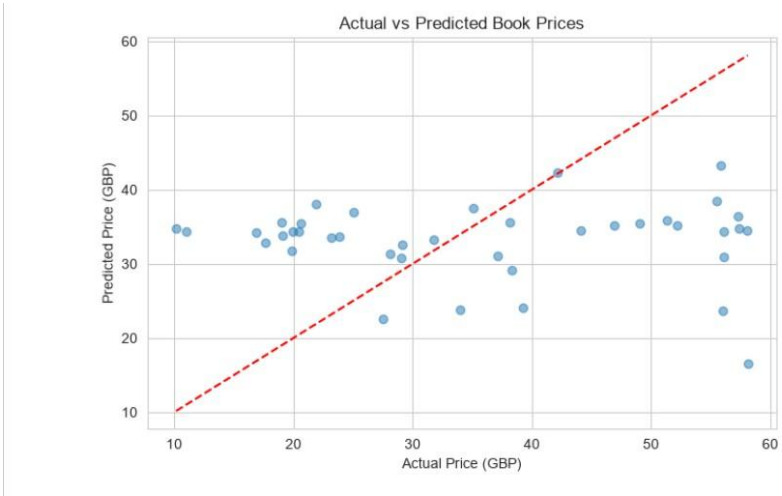
3.1.6 Figure 6

Impact of Stock Availability on Book Price. This boxplot compares the price distributions of books that are "In stock" versus "Out of stock", showing minimal price variation based on stock status.



3.1.7 Figure 7

Actual vs Predicted Book Prices (Linear Regression). This scatter plot compares the actual prices against the model's predicted prices. The red dashed line represents perfect predictions. The wide scatter indicates the model struggles to accurately estimate price.



3.2 Model Performance

The Linear Regression model produced the following evaluation metrics on the test set:

Metric	Value	Interpretation

Mean Absolute Error (MAE)	£14.01	Predictions are on average £14.01 away from actual price
Root Mean Squared Error (RMSE)	£16.46	Larger errors are penalised more heavily
R^2 Score	-0.174	The model performs worse than simply predicting the mean price

The negative R^2 score indicates that the model does not explain the variance in book prices and is outperformed by a naïve baseline. This confirms the weak correlations observed during exploratory analysis — book price in this dataset is not strongly determined by rating, category, stock, or textual features.

Coefficient Interpretation

The coefficients with the largest absolute magnitudes were primarily categorical:

- category_Autobiography (−22.48) — Autobiographies tend to be cheaper than the baseline.
- category_Philosophy (−17.99) — Philosophy books are also cheaper.
- category_Christian (+17.40) — Christian titles are significantly more expensive.
- category_Science (+14.46) — Science books carry a higher price.

This shows that *category* is the dominant signal in the model, while numeric features have minimal influence.

Discussion

The results reveal a clear finding: within this sandbox catalogue, book price cannot be reliably predicted using only the features available on the product listing page. The negative R^2 indicates that the linear regression model is not suitable for this problem in its current form.

There are several plausible explanations:

1. Missing predictive features: Real pricing in bookstores is influenced by author popularity, publisher, page count, publication date, format (hardcover vs paperback), and market demand — none of which are available on Books to Scrape.
2. Sandbox nature of data: Books to Scrape is a fictional catalogue designed for practising web scraping, not for meaningful market analysis. Prices may be randomly assigned.
3. Linear assumption violated: The relationship between features and price may be nonlinear, requiring more flexible models (e.g., Random Forest, Gradient Boosting).
4. Weak signal in features: Numeric features such as rating and description length have near-zero correlation with price.

Despite the weak predictive power, the project demonstrates a full and correct data science workflow: acquisition, cleaning, exploration, modelling, evaluation, and interpretation. The negative R^2 is itself an important, honest finding — not every model succeeds, and correctly identifying its limitations is part of good data science practice.

Conclusion

This project delivered a complete end-to-end data science pipeline for the Books to Scrape catalogue. 200 books were scraped, cleaned, visualized, and used to train a linear regression model. Despite the model's effective construction and evaluation, its predictive power was inadequate, indicating that the catalog lacks sufficient signal to estimate pricing based solely on product-page attributes.

Limitations

- The dataset is from a sandbox website and does not reflect real market behaviour.
- Only 200 books were used; a larger sample may improve generalisability.
- Linear regression assumes linear relationships, which the data does not appear to follow.
- High-cardinality categorical encoding may cause overfitting, especially with small data.

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